

UPEF répétitoire QCM Neurologie

2.06.2015

Frédéric Assal



L'ESSENTIEL, C'EST VOUS.



**UNIVERSITÉ
DE GENÈVE**

Un homme de 50 ans vient aux urgences, il se plaint de céphalées. Juste avant, il dit avoir vu des « choses brillantes » qui bougeaient. Il vous demande de fermer les stores. A quoi pensez-vous?

- a) Une hémorragie sous-arachnoïdienne
- b) Une migraine
- c) Une céphalée en grappe
- d) Une crise hypertensive
- e) Une crise d'épilepsie

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- a) Une hémorragie sous-arachnoïdienne
- b) Une migraine (avec aura visuelle)**
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- d) Une crise hypertensive
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IHS Classification ICHD-II



http://ihs-classification.org/en/02_klassifikation/02_teil1/01.00_00_migraine.html

Migraine

- A. Au moins 5 crises remplissant les critères de B à E
- B. Durée de 4 à 72 heures (sans traitement)
- C. Au moins 2 des caractéristiques suivantes
 - 1. céphalée unilatérale
 - 2. céphalée pulsatile
 - 3. modérée ou sévère
 - 4. aggravation par les activités physiques simples (montée escalier p.ex.)
- D. Durant les céphalées, au moins 1 des caractères suivants:
 - 1. nausées et/ou vomissements
 - 2. photophobie et phonophobie
- E. Non attribuable à une autre affection

Migraine avec aura typique

- A. Au moins 2 crises remplissant les critères B à E
- B. Symptômes visuels, sensitifs, dysphasiques mais pas de parésie
- C. Présence d'au moins 1 des symptômes suivants:
 - 1. symptômes visuels homonymes positifs (scotomes, lignes ou phosphènes scintillants) et/ou négatifs (amaurose, hémianopsie) et/ou symptômes sensitifs unilatéraux positifs (fourmillements, picotements) et/ou négatifs (anesthésie, engourdissement)
 - 2. ≥ 1 symptôme progresse sur ≥ 5 min ou différents symptômes se succèdent
 - 3. chaque symptôme dure 5-60 min
- D. La céphalée débute pendant l'aura ou lui succède dans les 60 min
- E. Non attribuable à une autre affection

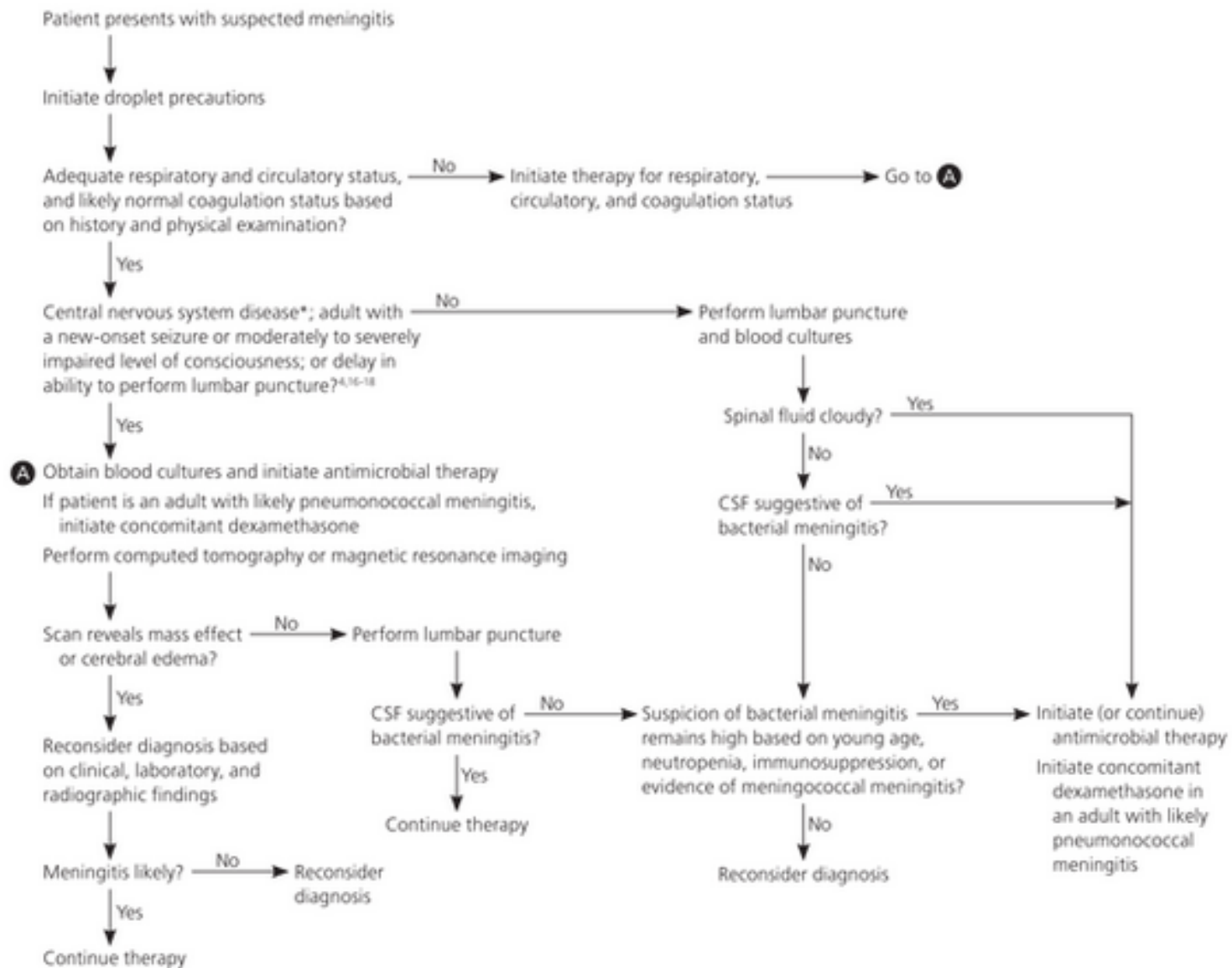
Migraine basilaire

Mêmes critères que la migraine avec aura et au moins 2 des symptômes suivants: diminution du champ visuel, bitemporal et binasal / dysarthrie / vertiges / tinnitus / diminution de l'acuité auditive / diplopie / ataxie / paresthésies bilatérales / parésies bilatérales / diminution du niveau de conscience

Quel est le signe associé à la méningite?

- a) **Le signe de Kernig (ext jambe>méninges)**
- b) Le signe de Tinel (t carpien)
- c) Le signe de Homans (TVP)
- d) Le signe de Babinski (pyramidal)
- e) Le signe de Richemond (inventé)

Initial Management of Suspected Acute Meningitis



*—Includes CSF shunts, hydrocephalus, trauma, space-occupying lesions or recent neurosurgery, immunocompromised state, papilledema, or focal neurological signs.

Dans quelle affection, pouvez-vous rencontrer des mouvements choréiques?

- a) La maladie de Parkinson
- b) La sclérose latérale amyotrophique
- c) La maladie de Gilles de la Tourette
- d) Le rhumatisme articulaire aigu; 1-8 mois après l'infection initiale (chorée de Sydenham) <http://www.uptodate.com/contents/sydenham-chorea>**
- e) La sclérose en plaques

Une atteinte du lobe frontal peut toucher les fonctions suivantes, sauf:

- a) Motricité fine (M1)
- b) Programmation-organisation (PF DL)
- c) Motivation (PF M)
- d) Expression du langage (Broca)
- e) **Compréhension du langage (Wernicke)**

The cortical language circuit: from auditory perception to sentence comprehension

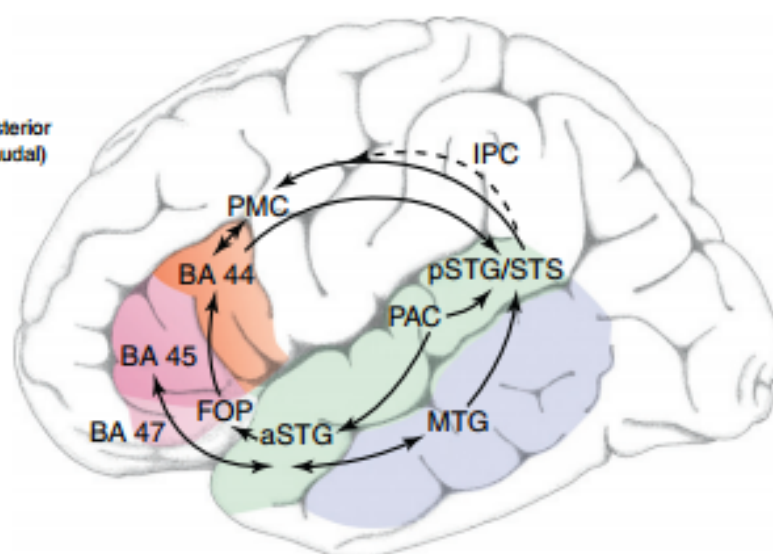
Opinio

Angela D. Friederici

16, No. 5

Max Planck Institute for Human Cognitive and Brain Sciences, Department of Neuropsychology, 04103 Leipzig, Germany

Superior
(dorsal)
Anterior
(rostral) ← Posterior
(caudal)
Inferior
(ventral)



IFG = Inferior frontal gyrus
STG = Superior temporal gyrus
MTG = Middle temporal gyrus

PAC = Primary auditory cortex
FOP = Frontal operculum
BA 44 = Pars opercularis
BA 45 = Pars triangularis
BA 47 = Pars orbitalis
PMC = Premotor cortex
IPC = Inferior parietal cortex

TRENDS in Cognitive Sciences

Figure 1. The cortical language circuit (schematic view of the left hemisphere). The major gyri involved in language processing are color-coded. In the frontal cortex, four language-related regions are labeled: three cytoarchitecturally defined Brodmann [39] areas (BA 47, 45, 44), the premotor cortex (PMC) and the ventrally located frontal operculum (FOP). In the temporal and parietal cortex the following regions are labeled: the primary auditory cortex (PAC), the anterior (a) and posterior (p) portions of the superior temporal gyrus (STG) and sulcus (STS), the middle temporal gyrus (MTG) and the inferior parietal cortex (IPC). The solid black lines schematically indicate the direct pathways between these regions. The broken black line indicates an indirect connection between the pSTG/STS and the PMC mediated by the IPC. The arrows indicate the assumed major direction of the information flow between these regions. During auditory sentence comprehension, information flow starts from PAC and proceeds from there to the anterior STG and via ventral connections to the frontal cortex. Back-projections from BA 45 to anterior STG and MTG via ventral connections are assumed to support top-down processes in the semantic domain, and the dorsal back-projection from BA 44 to posterior STG/STS to subserve top-down processes relevant for the assignment of grammatical relations. The dorsal pathway from PAC via pSTG/STS to the PMC is assumed to support auditory-to-motor mapping. Furthermore, within the temporal cortex, anterior and posterior regions are connected via the inferior and middle longitudinal fasciculi, branches of which may allow information flow from and to the mid-MTG.

Quand quelqu'un ne peut pas reconnaître un objet mais le dénomme correctement, on parle de:

- a) Prosopagnosie (non rec pers familiers)
- b) Anomie (défaut du mot)
- c) Aphasie agnosique (?)
- d) Agnosie (visuelle)**
- e) Anosognosie (méconnaissance du trouble)

La maladie de la vache folle est:

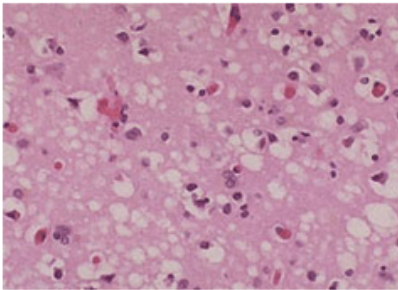
- a) Une méningite spongiforme
- b) Une spirilose spongiforme
- c) Une méningo-encéphalite
- d) Une encéphalopathie spongiforme (PrP)**
- e) Une encéphalomyélite

Human Prion Diseases

- Creutzfeldt-Jakob Disease (CJD)
- Variant Creutzfeldt-Jakob Disease (vCJD)
- Gerstmann-Straussler-Scheinker Syndrome
- Fatal Familial Insomnia
- Kuru

Animal Prion Diseases

- Bovine Spongiform Encephalopathy (BSE)
- Chronic Wasting Disease (CWD)
- Scrapie
- Transmissible mink encephalopathy
- Feline spongiform encephalopathy
- Ungulate spongiform encephalopathy



CJD (CREUTZFELDT-JAKOB DISEASE, CLASSIC)

Classic CJD is a human prion disease. It is a neurodegenerative disorder with characteristic clinical and diagnostic features.

[More >](#)



VCJD (VARIANT CREUTZFELDT-JAKOB DISEASE)

vCJD has a different clinical and pathologic characteristics from classic CJD. Each disease also has a particular genetic profile of the prion protein gene.

[More >](#)



BSE (BOVINE SPONGIFORM ENCEPHALOPATHY)

BSE also known as Mad Cow Disease is a progressive neurological disorder of cattle that results from infection by an unusual transmissible agent called a prion.

[More >](#)



CWD (CHRONIC WASTING DISEASE)

CWD is a prion disease that affects North American cervids (hoofed ruminant mammals, with males characteristically having antlers).

[More >](#)

Après un impact crânien, quelles sont les complications possibles, sauf?

- a) Une commotion cérébrale
- b) Une encéphalite**
- c) Une brèche durale
- d) Une dissection vertébrale
- e) Une contusion cérébrale

<http://www.uptodate.com/contents/traumatic-brain-injury-epidemiology-classification-and-pathophysiology>

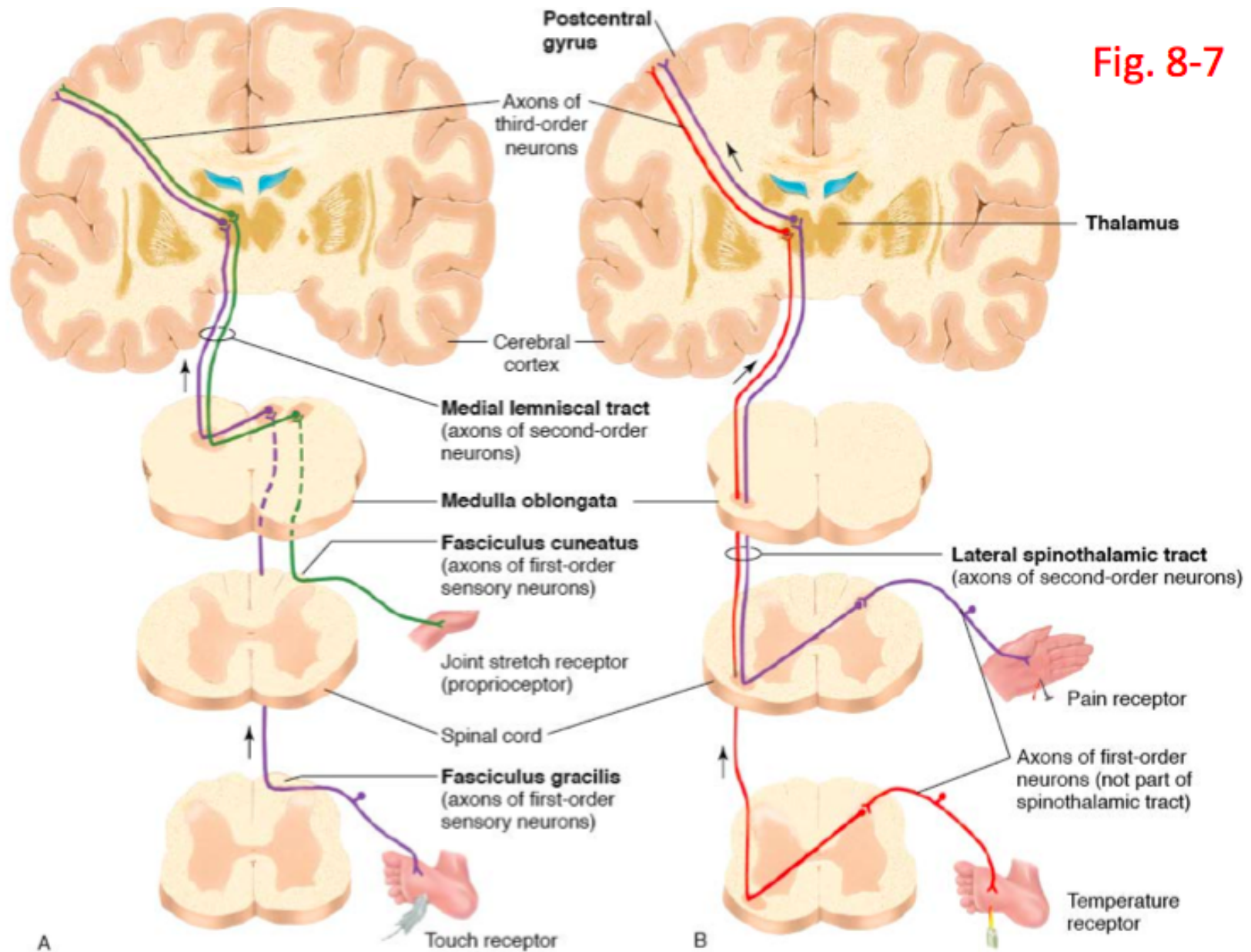
En cas d'atteinte du motoneurone supérieur, on peut mettre en évidence:

- a) **Une spasticité**
- b) **Une roue dentée**
- c) **Des myoclonies**
- d) **Une paratonie**
- e) **Une ataxie**

Une diminution ou abolition de la sensibilité thermoalgique d'origine médullaire se voit dans:

- a) Atteinte des cornes antérieures
- b) Atteinte des cordons postérieurs
- c) Syndrome de la queue de cheval
- d) Atteinte centromédullaire**
- e) Atteinte des cordons antérolatéraux

Fig. 8-7



Une hypoesthésie en gant et chaussettes se voit généralement dans les atteintes:

- a) De la moelle
- b) Du thalamus
- c) D'une ou de plusieurs racines
- d) Du cortex pariétal primaire
- e) Du nerf périphérique**

Diabetic neuropathy: clinical manifestations and current treatments



Brian C Callaghan, Hsinlin T Cheng, Catherine L Stables, Andrea L Smith, Eva L Feldman

Diabetic peripheral neuropathy is a prevalent, disabling disorder. The most common manifestation is distal symmetrical polyneuropathy (DSP), but many patterns of nerve injury can occur. Currently, the only effective treatments are glucose control and pain management. While glucose control substantially decreases the development of neuropathy in those with type 1 diabetes, the effect is probably much smaller in those with type 2 diabetes. Evidence supports the use of specific anticonvulsants and antidepressants for pain management in patients with diabetic peripheral neuropathy. However, the lack of disease-modifying therapies for diabetic DSP makes the identification of new modifiable risk factors essential. Growing evidence supports an association between components of the metabolic syndrome, including prediabetes, and neuropathy. Studies are needed to further explore this association, which has implications for the development of new treatments for this common disorder.

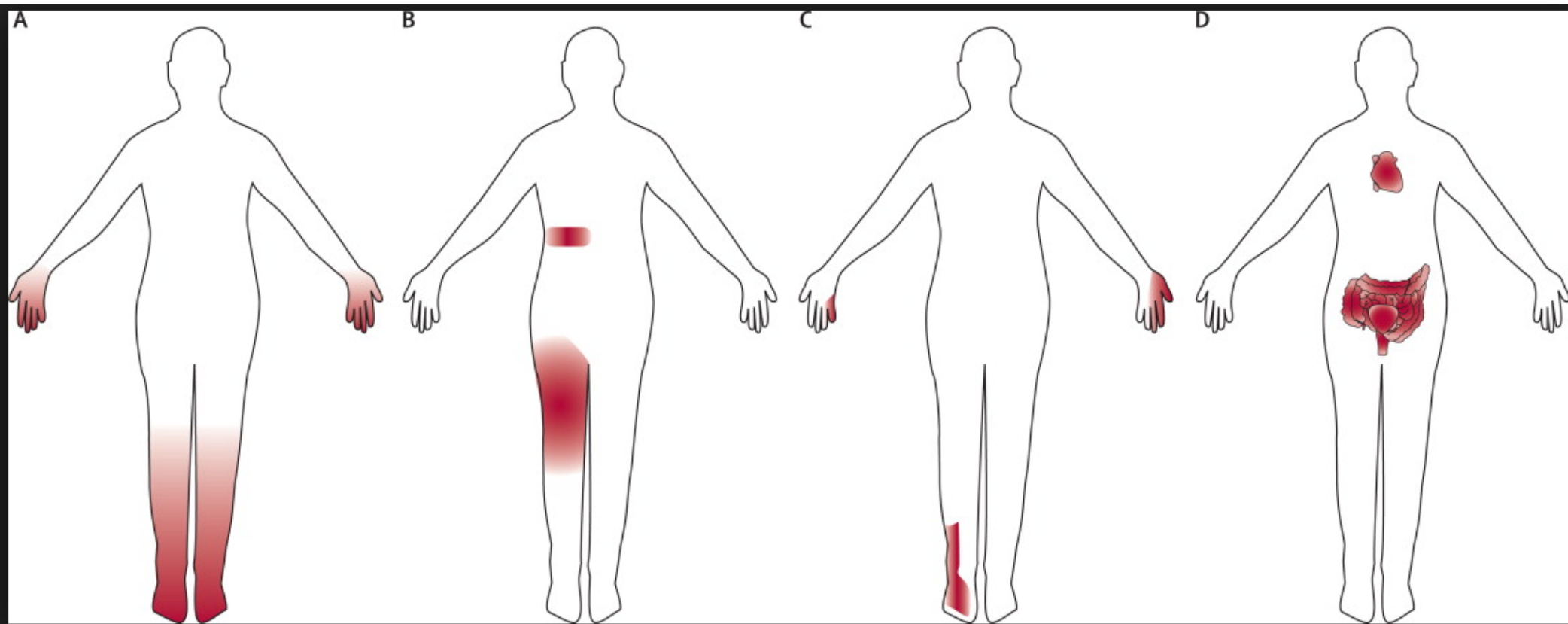
Lancet Neurol 2012; 11: 521-34

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L'incapacité à marcher droit sur une ligne se rencontre dans les affections:

- a) **Du cervelet**
- b) Du lobe pariétal
- c) Du lobe frontal
- d) Du corps calleux
- e) Du lobe occipital

SHORT REPORT

Typical features of cerebellar ataxic gait

H Stolze, S Klebe, G Petersen, J Raethjen, R Wenzelburger, K Witt, G Deuschl

J Neurol Neurosurg Psychiatry 2002;73:310-312

Background: Although gait disturbance is one of the most pronounced and disabling symptoms in cerebellar disease (CD), quantitative studies on this topic are rare.

Objectives: To characterise the typical clinical features of cerebellar gait and to analyse ataxia quantitatively.

Methods: Twelve patients with various cerebellar disorders were compared with 12 age matched controls. Gait was analysed on a motor driven treadmill using a three dimensional system. A tandem gait paradigm was used to quantify gait ataxia.

Results: For normal locomotion, a significantly reduced step frequency with a prolonged stance and double limb support duration was found in patients with CD. All gait measurements were highly variable in CD. Most importantly, balance related gait variables such as step width and foot rotation angles were increased in CD, indicating the need for stability during locomotion. The tandem gait paradigm showed typical features of cerebellar ataxia such as dysmetria, hypometria, hypermetria, and inappropriate timing of foot placement.

Conclusions: Typical features of gait in CD are reduced cadence with increased balance related variables and an almost normal range of motion (with increased variability) in the joints of the lower extremity. The tandem gait paradigm accentuates all the features of gait ataxia and is the most sensitive clinical test.

a hindrance or disease that interfered with an unrestrained gait, other than of cerebellar origin (for the patients). All subjects gave their informed consent to participate in the study, which was approved by the local ethics committee.

Gait analysis was carried out for two different tasks: (a) normal locomotion; (b) tandem gait. The detailed method used to calculate the various gait variables, during both normal locomotion and tandem gait, has been described elsewhere.⁴ The natural walking speed, during normal locomotion and tandem gait, of each subject was measured during overground locomotion and was then preselected on the treadmill. During tandem gait, the subjects were instructed to place one foot in front of the other on a line of red tape that was attached to the lamella layers of the treadmill. Before recording, the subjects were given five minutes to familiarise themselves with treadmill locomotion. Gait was recorded with a three dimensional infrared movement analysis system (Qualisys, Sandviken, Sweden). Seven infrared light reflective markers were attached to each leg. For each patient, 15-20 consecutive walking cycles were averaged for calculation of the different spatiotemporal gait measurements, including kinematics, with self developed software.⁵ During tandem gait, an ataxia ratio measuring the regularity of the strides in all three room directions was calculated as a ratio of the standard deviation (SD) of foot placement: (SD of step length + SD of step width + SD of step height)/3. Missteps, defined as steps not hitting the tape, were counted over 60 seconds during tandem gait. For statistical analysis, the two groups were compared using the Mann-Whitney U

Dans les atteintes du cervelet, l'examen peut montrer, sauf:

- a) Une atteinte au test doigt-nez
- b) Un test de Weber latéralisé**
- c) Une atteinte au test talon-genou
- d) Une manœuvre de Romberg positive
- e) Une hypotonie

16. Savoir interpréter l'épreuve de Rinne et l'épreuve de Weber ; distinguer surdité de transmission, de perception.



L'acoumétrie est un temps essentiel dans l'approche d'une hypoacousie. C'est le dernier temps de l'examen clinique en otologie. Il doit être effectué :

- Une acoumétrie à la voix chuchotée : permet de suspecter une hypoacousie.
- Une acoumétrie au diapason : plus fiable, utilisant des tests simples permettant d'orienter le praticien vers un diagnostic qualitatif de la surdité : perception ou transmission. Les tests les plus utilisés sont :

• L'épreuve de RINNE :

Le diapason est mis en vibration. Son pied d'abord appliqué sur la mastoïde (Conduction Osseuse Relative = COR). Lorsqu'il n'est plus entendu, le diapason est présenté à 10 cm du pavillon (Conduction Aérienne = CA).

Les résultats sont présentés dans le tableau suivant :

Sujet normal	Surdité de perception	Surdité de transmission
Le son est réentendu	Le son est réentendu	Le son n'est pas réentendu
CA>COR	CA>COR	CA=COR
Rinne positif ou nul	Rinne positif ou nul	Rinne négatif

• L'épreuve de WEBER :

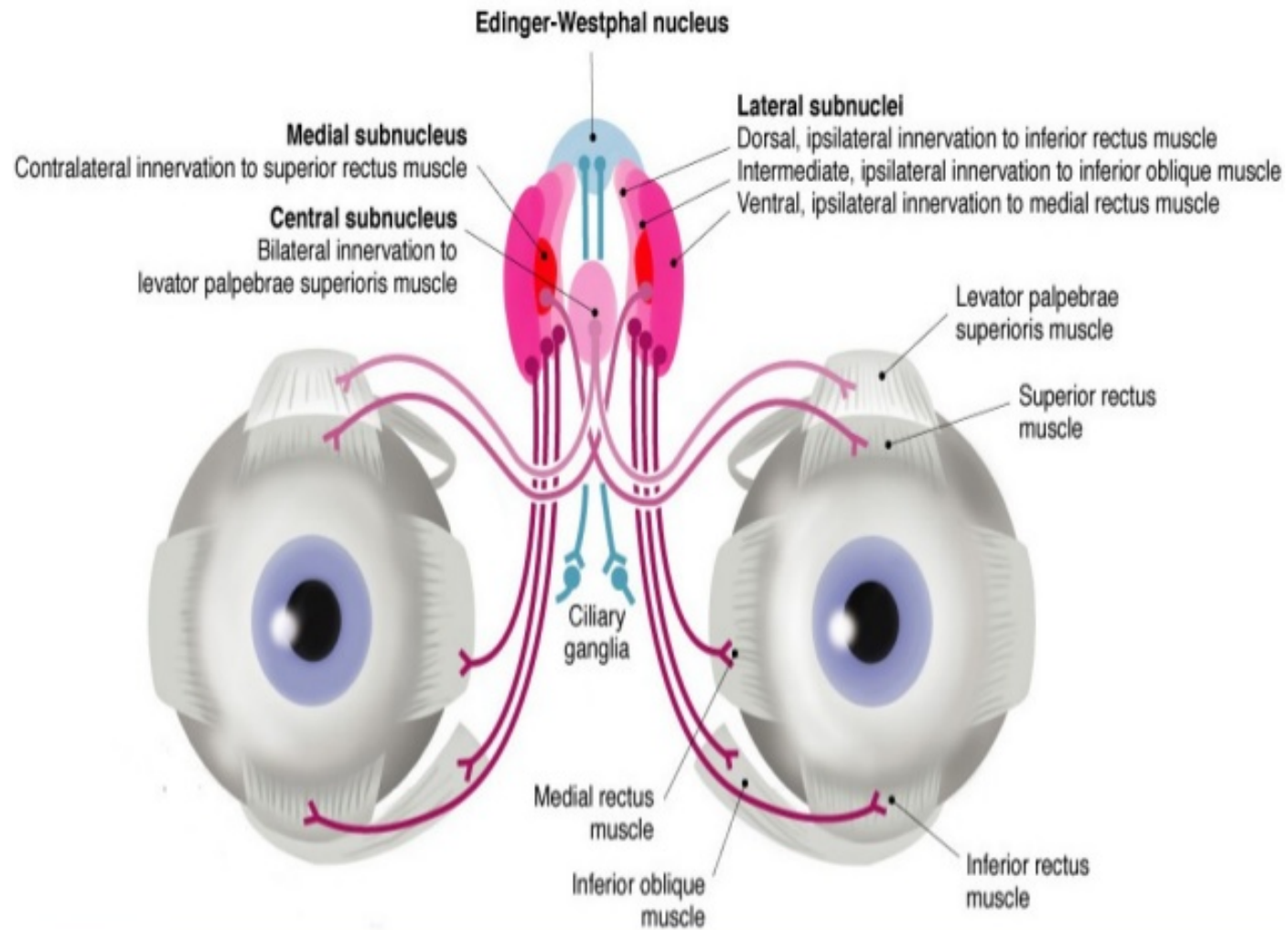
Le diapason est appliqué sur le vertex ou sur le front par son pied. Le sujet perçoit les vibrations transmises sous forme d'un bourdonnement.

Les résultats obtenues sont illustrés dans le tableau suivant :

Sujet normal	Surdité de perception	Surdité de transmission
Son perçu par les 2 oreilles Ou Le sujet ne peut pas préciser le côté (indifférent)	Weber latéralisé du côté sain	Weber latéralisé du côté malade

La présence d'une ptose palpébrale suggère une atteinte de quel NC:

- a) IV (diplopie vers le bas et du côté sain)
- b) VII (lagophtalmie)
- c) V3 (faiblesse des masticateurs)
- d) III (releveur de la paupière)**
- e) VI (diplopie regard externe)



Une atteinte des NC IX, X, XI et XII cause un:

- a) Syndrome de Brown-Séquard
- b) Syndrome pseudo-bulbaire
- c) Syndrome bulbaire**
- d) Syndrome de Gertsman
- e) Syndrome de Parinaud

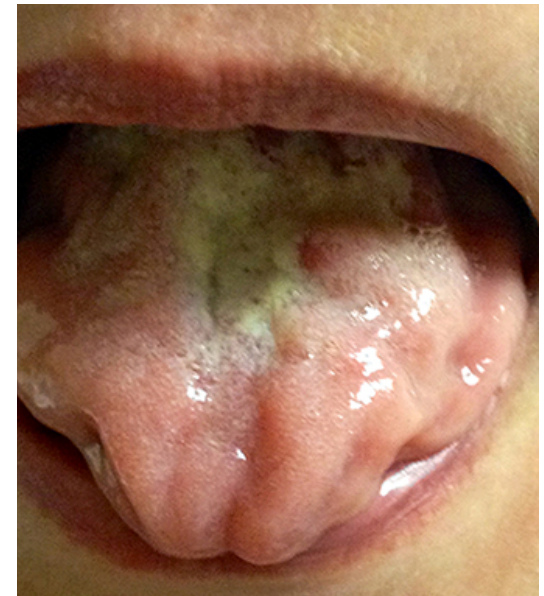
	Bulbar palsy	Pseudobulbar palsy	Opercular syndrome
Site of lesion	Nuclear/infranuclear (brain stem/nerves/neuromuscular junction/muscle diseases)	Supranuclear (upper motor neuron) (bilateral cortico-bulbar pathways)	"Cortical variant" of supranuclear palsy (bilateral opercular cortex or its subcortical projections)
Weakness	Flaccid	Spastic	Present unclassifiable
Atrophy	++	-	Absent
Fasciculations	++	-	Absent
Dysphagia	++ Severe, nasal regurgitation	Mild, Coughing more than nasal regurgitation	Moderate, Coughing as well as nasal regurgitation
Speech	Nasal, breathy,	Heavy, spastic	Mixed spastic and nasal
Automatic movements	Absent	Absent	Present
Facial weakness	LMN type	If present, of UMN type	Difficult to classify
Palatal gag reflex	Absent	Exaggerated	Absent
Jaw jerk	Absent	Exaggerated	Normal/absent
Other release signs: Snout/ suck/frontal release signs	Absent	Present	Absent
Emotional incontinence	Absent	Present	Absent
Long track UMN signs in limbs	Present if site in brain stem absent if site is extra-axial nerve, or distal	Present	Present/absent

Des fasciculations se rencontrent dans l'atteinte de:

- a) Cordons postérieurs de la moelle
- b) Capsule interne
- c) Cortex moteur
- d) Corne antérieure de la moelle (SLA)**
- e) Cordons antérolatéraux de la moelle

Fasciculations

- Brief discharges of individual axons
 - Cause contraction of muscle fibers in all, or part of, motor unit
 - Produce visible rippling of muscle
 - Origin
 - May originate
 - Locations: Cell body, anywhere along course of axon or axon terminal
 - Most often originate: Distal axon
 - Properties: 2 types
 - Axonal origin
 - May have short interspike intervals
 - Refractory period 3 to 4 msec
 - ISI 5 msec
 - Proximal (Cell body) origin
 - Refractory period 17 to 46 msec
 - ISI 80 msec
 - EMG appearance: Single motor unit
 - Onset
 - Often spontaneous
 - May be precipitated by: Muscle percussion or contraction, fatigue, cold
 - Discharge frequency & patterns
 - Rhythm: Irregular
 - Size & Shape: Same as motor unit potentials
 - Rate & Disease association
 - Benign
 - Short interval between discharges (0.5s to 1s)
 - More common in gastrocnemius
 - Motor neuron disease³⁷
 - Longer interval between discharges (3.5s)
 - Fasciculation frequency not related to disease duration or weakness
 - Accompanied by fibrillations
 - Grouped occurrence from multiple units
 - More common with strength deterioration
 - Related to cortical hyperexcitability
 - Common in: Proximal muscles with some weakness
 - Difference from Voluntary activity (Motor unit potentials)
 - Begin firing at 4 to 8 Hz: Differs among muscles



Quel est le mouvement anormal le plus fréquent dans la maladie de Parkinson débutante:

- a) La chorée (Huntington ou forme évol ss l-dopa)
- b) Les dyskinésies (sous l-dopa)
- c) La dystonie (dég cortico-basale ou forme évol ss l-dopa)
- d) Les myoclonies (mal c de Lewy)
- e) Le tremblement (de repos)**

Table 3. The UK Parkinson's Disease Society Brain Bank clinical diagnostic criteria

Step 1: Diagnosis of parkinsonian syndrome

Bradykinesia (slowness of initiation of voluntary movement with progressive reduction in speed and amplitude or repetitive actions)

And at least one of the following:

Muscular rigidity

4–6 Hz rest tremor

Postural instability not caused by primary visual, vestibular, cerebellar, or proprioceptive dysfunction

Step 2: Exclusion criteria for Parkinson's disease

History of repeated strokes with stepwise progression of parkinsonian features

History of repeated head injury

History of definite encephalitis

Oculogyric crises

Neuroleptic treatment at onset of symptoms

More than one affected relative^a

Sustained remission

Strictly unilateral features after 3 years

Supranuclear gaze palsy

Cerebellar signs

Early severe autonomic involvement

Early severe dementia with disturbances of memory, language, and praxis

Babinski sign

Presence of a cerebral tumor or communicating hydrocephalus on CT scan

Negative response to large doses of levodopa (if malabsorption excluded)

MPTP exposure

Step 3: Supportive positive criteria of Parkinson's disease

Three or more required for diagnosis of definite Parkinson's disease:

Unilateral onset

Rest tremor present

Progressive disorder

Persistent asymmetry affecting the side onset most

Excellent response (70%–100%) to L-dopa

Severe levodopa-induced chorea

Levodopa response for 5 years or more

Clinical course of 10 years or more

Hyposmia

Visual hallucinations

^aThis criterion is no longer used (Hughes et al. 1992; Litvan et al. 2003; Lees et al. 2009).

Cold Spring Harb Perspect Med 2012;2:a008870

Une atteinte du nerf trijumeau entraine:

- a) Une parésie de l'hémiface (PF)
- b) Un trouble de la déglutition
- c) Une hémicrânie (migraine ou cluster)
- d) Une douleur paroxystique de l'hémiface
(V2 > V1 > V3)**
- e) Une lagophtalmie (PF périph)

IHS Diagnostic Criteria for Trigeminal Neuralgia

Classical

A. Paroxysmal attacks of pain lasting from a fraction of a second to two minutes, affecting one or more divisions of the trigeminal nerve, and fulfilling criteria B and C

B. Pain has at least one of the following characteristics:

1. Intense, sharp, superficial, or stabbing
2. Precipitated from trigger zones or by trigger factors

C. Attacks are stereotyped in the individual patient

D. There is no clinically evident neurologic deficit

E. Not attributed to another disorder

Symptomatic

A. Paroxysmal attacks of pain lasting from a fraction of a second to two minutes, with or without persistence of aching between paroxysms, affecting one or more divisions of the trigeminal nerve, and fulfilling criteria B and C

B. Pain has at least one of the following characteristics:

1. Intense, sharp, superficial, or stabbing
2. Precipitated from trigger zones or by trigger factors

C. Attacks are stereotyped in the individual patient

D. A causative lesion, other than vascular compression, has been demonstrated by special investigations and/or posterior fossa exploration

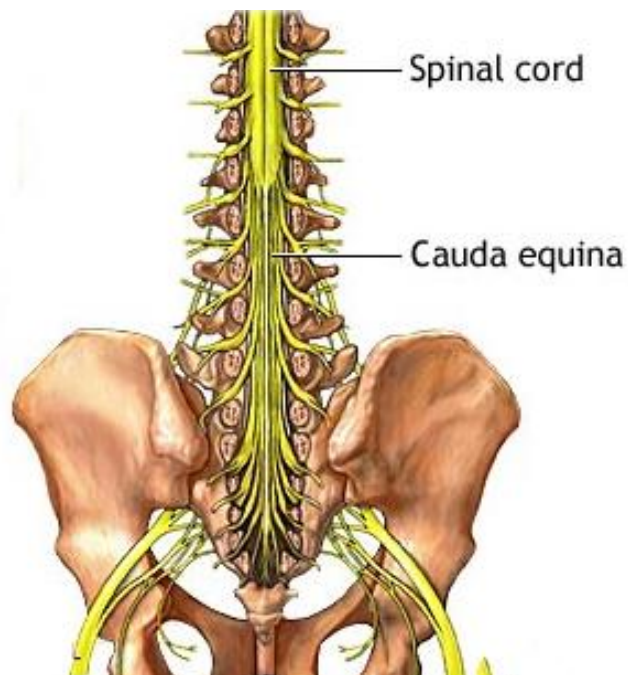
IHS = International Headache Society.

Information from reference 1.

<http://www.aafp.org/afp/2008/0501/p1291.html>

Dans le syndrome de la queue de cheval, on peut rencontrer, sauf:

- a) Des douleurs des membres inférieurs
- b) Une anesthésie des membres inférieurs
- c) Une anesthésie du périnée
- d) Une paraparésie spastique**
- e) Une paraparésie flasque

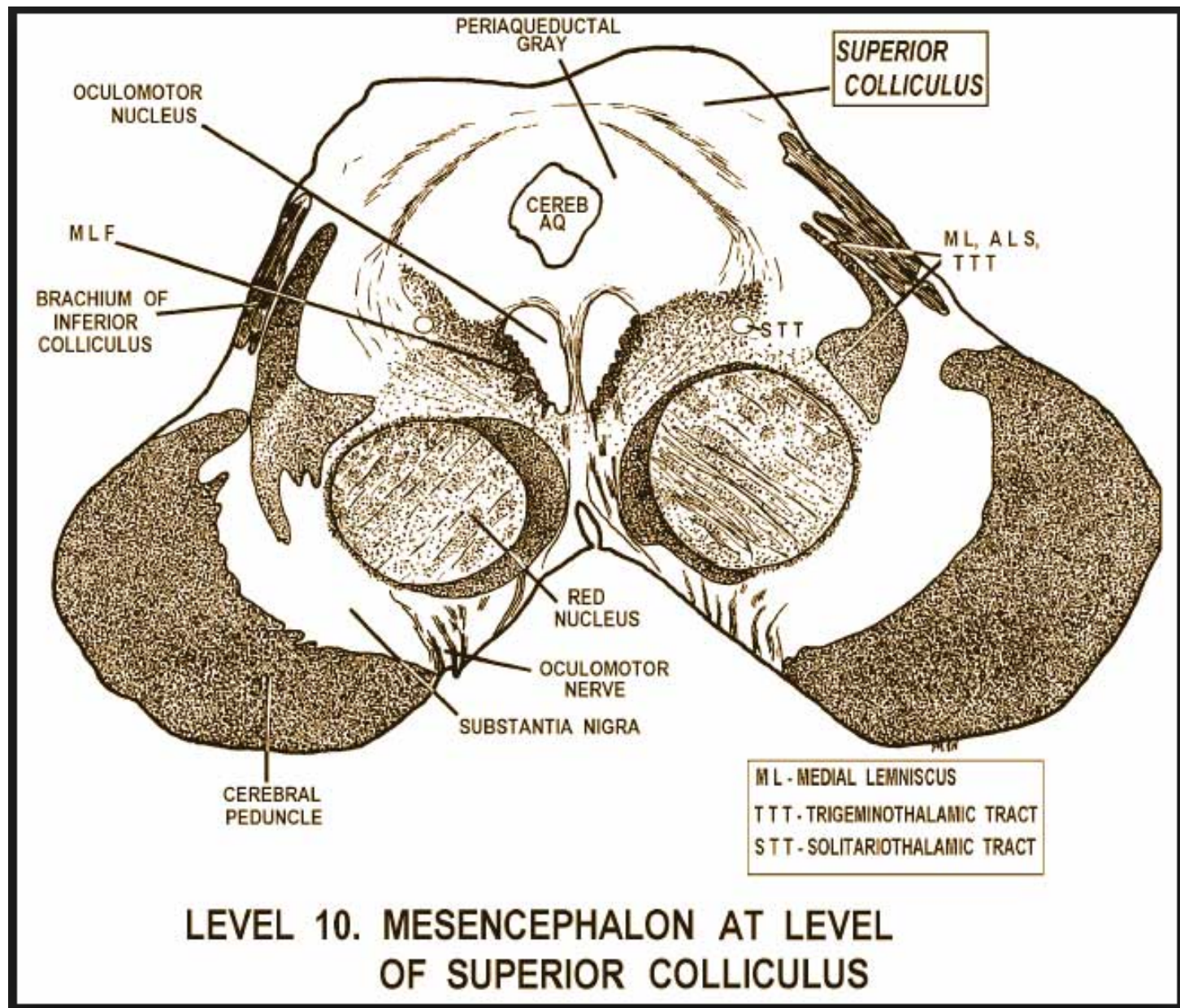


Conus Medullaris vs. Cauda Equina Syndromes

	Conus medullaris syndrome	Cauda equina syndrome
Vertebral level	L1-L2	L2-sacrum
Spinal level	Sacral cord segment and roots	Lumbosacral nerve roots
Presentation	Sudden and bilateral	Gradual and unilateral
Radicular pain	Less severe	More severe
Low back pain	More	Less
Motor strength	Symmetrical, less marked hyperreflexic distal paresis of LL, fasciculation	More marked asymmetric areflexic paraplegia, atrophy more common
Reflexes	Ankle jerks affected	Both knee and ankle jerks affected
Sensory	Localized numbness to perianal area, symmetrical and bilateral	Localized numbness at saddle area, asymmetrical, unilateral
Sphincter dysfunction	Early urinary and fecal incontinence	Tend to present late
Impotence	Frequent	Less frequent

L'atteinte du mésencéphale peut provoquer, sauf:

- a) Une hémiplégie (tr pyramidal)
- b) Une parésie oculomotrice (III ou IV)
- c) Une atrophie de la langue (bulbe)**
- d) Une ptose (III)
- e) Un tremblement (« midbrain tremor »)



<http://www.neuroanatomy.wisc.edu/virtualbrain/BrainStem/23Colliculus.html>

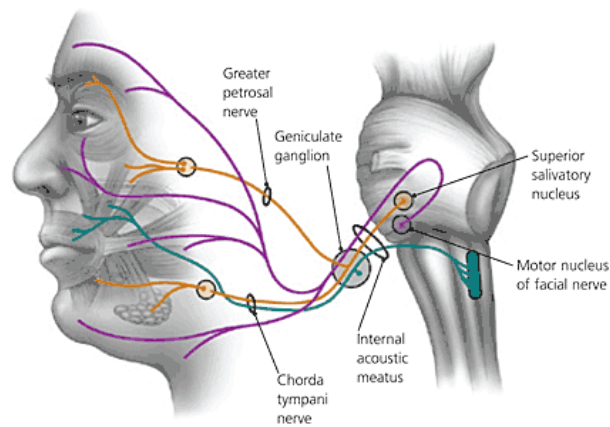
Une paralysie faciale périphérique « a frigore » peut entraîner, sauf:

- a) Une lagophtalmie
- b) Une hyperacousie
- c) Une dysgueusie
- d) Une hypoacousie**
- e) Un trouble sensitif subjectif

Differential Diagnosis for Facial Nerve Palsy

DISEASE	CAUSE	DISTINGUISHING FACTORS
Nuclear (peripheral)		
Lyme disease	Spirochete <i>Borrelia burgdorferi</i>	History of tick exposure, rash, or arthralgias; exposure to areas where Lyme disease is endemic
Otitis media	Bacterial pathogens	Gradual onset; ear pain, fever, and conductive hearing loss
Ramsay Hunt syndrome	Herpes zoster virus	Pronounced prodrome of pain; vesicular eruption in ear canal or pharynx
Sarcoidosis or Guillain-Barré syndrome	Autoimmune response	More often bilateral
Tumor	Cholesteatoma, parotid gland	Gradual onset
Supranuclear (central)		
Multiple sclerosis	Demyelination	Forehead spared
Stroke	Ischemia, hemorrhage	Extremities on affected side often involved
Tumor	Metastases, primary brain	Gradual onset; mental status changes; history of cancer

- Visceral efferent fibers (facial expression muscles, stapedius muscle)
- Visceral motor fibers (lacrimal, salivary glands)
- Special sensory fibers (supplies taste to anterior two thirds of the tongue)



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Les signes/symptômes de l'hypertension intracrânienne sont, sauf:

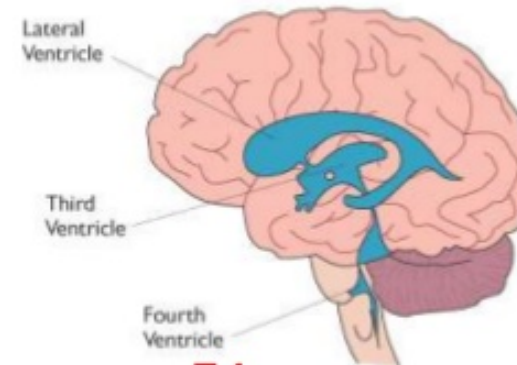
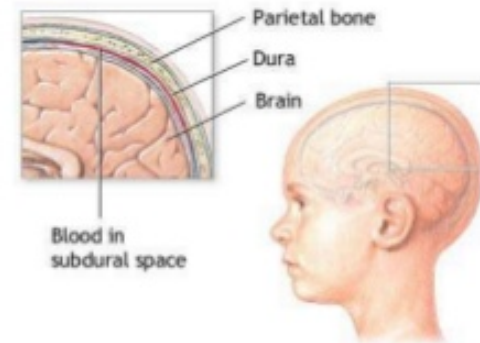
- a) Des céphalées matinales
- b) Des nausées et vomissements
- c) Une bradycardie
- d) Un oedème papillaire
- e) Des attaques de panique**

Increased Intracranial Pressure

✓ Causes

- ✓ Tumors
- ✓ Accumulation of fluid within the ventricular system
- ✓ Bleeding
- ✓ Edema in cerebral tissues

✓ Early signs and symptoms are often subtle and assume many patterns



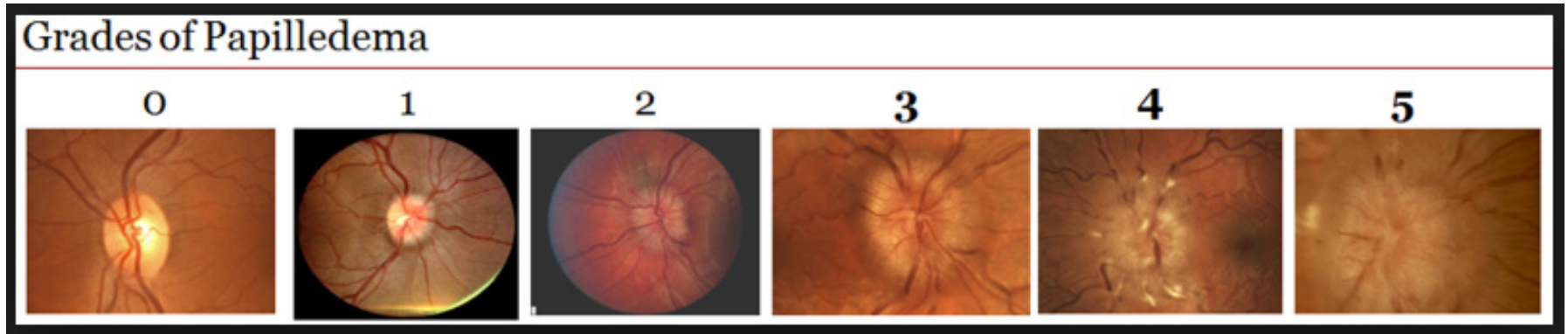
54

Dans la névrite optique rétrobulbaire, il n'a pas de:

- a) Pâleur de la papille au FO
- b) Pupille de Gunn
- c) Douleurs lors de mouvements oculaires
- d) Diminution de la vision des couleurs
- e) **Oedème papillaire**



http://www.nature.com/eye/journal/v25/n7/fig_tab/eye201181f3.html



<http://webeye.ophth.uiowa.edu/dept/iih/IIHTT/index.htm>

En cas d'épilepsie focale, quel est l'examen de première intention:

- a) EEG
- b) CT cérébral/IRM cérébrale**
- c) Examen neuropsychologique
- d) Fond d'œil
- e) Potentiels évoqués sensitifs

Adults who present to the emergency department after an unprovoked first seizure should receive immediate neuroimaging of the brain if feasible, although testing at a later date may be acceptable in certain patients if follow-up is reliable

American College of Emergency Physicians Clinical Policies Committee, Clinical Policies Subcommittee on Seizures. Clinical policy: critical issues in the evaluation and management of adult patients presenting to the emergency department with seizures. *Ann Emerg Med*. 2004;43:605-25

<http://www.aafp.org/afp/2007/0501/p1342.html>

Si vous suspectez une épilepsie-absence de l'enfant, quel est l'examen de choix:

- a) **EEG (bouffées 3 c/seconde)**
- b) IRM cérébrale (normale)
- c) Examen neuropsychologique (normal)
- d) Fond d'œil (normal)
- e) CT-cérébral (RX!)

EEG IN THE DIAGNOSIS, CLASSIFICATION, AND MANAGEMENT OF PATIENTS WITH EPILEPSY

S J M Smith

J Neurol Neurosurg Psychiatry 2005;76(Suppl II):ii2-ii7. doi: 10.1136/jnnp.2005.069245

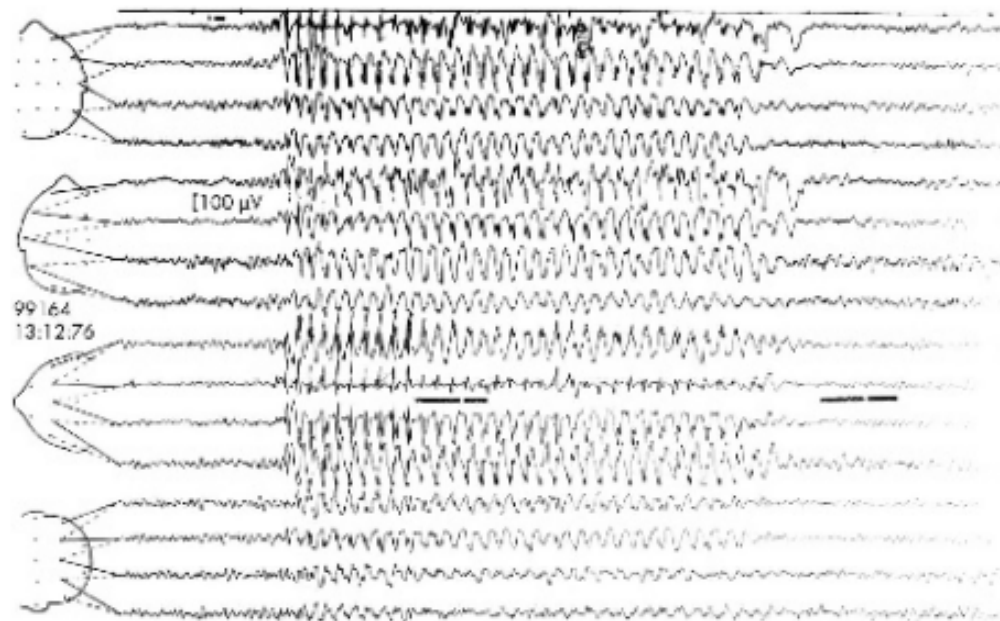


Figure 1 Three per second spike and wave discharge during typical absence seizure.

Le syndrome de Korsakoff, souvent rencontré chez l'alcoolique est attribué à :

- a) Une carence en vitamine B12
- b) Un effet toxique de l'alcool
- c) Une carence en vitamine B1**
- d) Une carence en vitamine E
- e) Une hypoxie cérébrale

The Korsakoff Syndrome: Clinical Aspects, Psychology and Treatment

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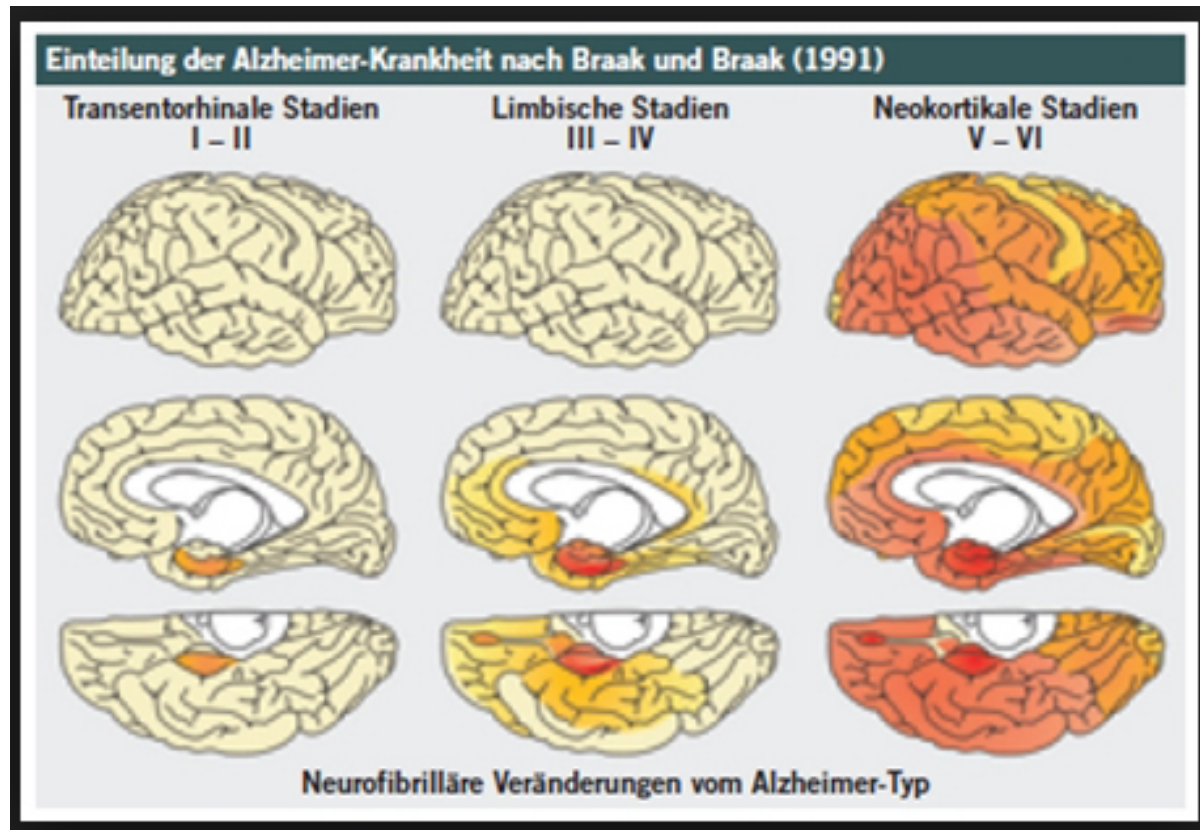
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Abstract — Aims: The Korsakoff syndrome is a preventable memory disorder that usually emerges (although not always) in the aftermath of an episode of Wernicke's encephalopathy. The present paper reviews the clinical and scientific literature on this disorder. **Methods:** A systematic review of the clinical and scientific literature on Wernicke's encephalopathy and the alcoholic Korsakoff syndrome. **Results:** The Korsakoff syndrome is most commonly associated with chronic alcohol misuse, and some heavy drinkers may have a genetic predisposition to developing the syndrome. The characteristic neuropathology includes neuronal loss, micro-haemorrhages and gliosis in the paraventricular and peri-aqueductal grey matter. Lesions in the mammillary bodies, the mammillo-thalamic tract and the anterior thalamus may be more important to memory dysfunction than lesions in the medial dorsal nucleus of the thalamus. Episodic memory is severely affected in the Korsakoff syndrome, and the learning of new semantic memories is variably affected. 'Implicit' aspects of memory are preserved. These patients are often first encountered in general hospital settings where they can occupy acute medical beds for lengthy periods. Abstinence is the cornerstone of any rehabilitation programme. Korsakoff patients are capable of new learning, particularly if they live in a calm and well-structured environment and if new information is cued. There are few long-term follow-up studies, but these patients are reported to have a normal life expectancy if they remain abstinent from alcohol. **Conclusions:** Although we now have substantial knowledge about the nature of this disorder, scientific questions (e.g. regarding the underlying genetics) remain. More particularly, there is a dearth of appropriate long-term care facilities for these patients, given that empirical research has shown that good practice has beneficial effects.

<http://alcalc.oxfordjournals.org/content/alcalc/44/2/148.full.pdf>

Quelle est la région la plus épargnée dans la maladie d'Alzheimer:

- a) Le lobe frontal
- b) Le lobe temporal
- c) Le lobe pariétal
- d) Le lobe occipital**
- e) L'hippocampe



<http://www.scilogs.de/die-sankore-schriften/fatale-folgen-f-r-sfpq-bei-alzheimer/>

Un homme de 50 ans, hypertendu, constate au réveil une fermeture incomplète de l'œil droit. A l'examen, il y a aussi une diminution du réflexe cornéen droit. Quel est le diagnostic le plus probable?

- a) Une myasthénie grave
- b) Un AVC ischémique
- c) Une paralysie faciale périphérique**
- d) Une sclérose en plaques
- e) Une maladie de Steinert

Devant une paralysie ascendante subaigue au décours de diarrhées et une aréflexie des membres inférieurs, quel est l'examen le plus contributif dans un premier temps?

- a) Une IRM cérébrale
- b) Une IRM lombaire
- c) Une IRM cervicodorsale
- d) Un ENMG
- e) Une ponction lombaire (GBS, dissoc AC)**

Box 1 | Diagnostic criteria for GBS^{2,4,6,7}**Features required for diagnosis of GBS***

- Progressive weakness in legs and arms (sometimes initially only in legs)
- Areflexia (or decreased tendon reflexes) in weak limbs

Acute inflammatory demyelinating polyneuropathy (AIDP)

Additional symptoms

- Progressive phase lasts days to 4 weeks
- Relative symmetry of symptoms
- Mild sensory symptoms or signs
- Cranial nerve involvement, especially bilateral weakness of facial muscles
- Autonomic dysfunction
- Pain (often)

Nerve conduction study findings

- Features of demyelination (only assessable if distal CMAP amplitude is >10% LLN)
- Prolonged distal motor latency
- Decreased motor nerve conduction velocity[‡]
- Increased F-wave latency, conduction blocks and temporal dispersion

Acute motor axonal neuropathy (AMAN)

Additional symptoms

- Progressive phase lasts days to 4 weeks
- Relative symmetry of symptoms
- No sensory symptoms or signs
- Cranial nerve involvement (rarely)
- Autonomic dysfunction
- Pain (sometimes)

Nerve conduction study findings

- No features of demyelination (or, one demyelinating feature in one nerve if distal CMAP amplitude is <10% LLN)
- Distal CMAP amplitude is <80% LLN in at least two nerves
- Transient motor nerve conduction block may be present (possibly caused by antiganglioside antibodies)

Features that should raise doubt about the diagnosis of GBS

- Increased number of mononuclear cells in cerebrospinal fluid (>50 cells per µl) or polymorphonuclear cells in cerebrospinal fluid
- Severe pulmonary dysfunction with limited limb weakness at onset
- Severe sensory signs with limited weakness at onset
- Bladder or bowel dysfunction at onset
- Fever at onset
- Sharp spinal cord sensory level
- Slow progression with limited weakness and without respiratory involvement (consider subacute inflammatory demyelinating polyneuropathy or acute-onset chronic inflammatory demyelinating polyneuropathy)
- Marked persistent asymmetry of weakness
- Persistent bladder or bowel dysfunction

*Classification of GBS as either AIDP or AMAN is not required for diagnosis of GBS, and whether AIDP and AMAN require different treatments is unknown. [‡]The amount of conduction slowing required to define demyelination differs between classification systems.^{6,23} Abbreviations: CMAP, compound muscle action potential; GBS, Guillain-Barré syndrome; LLN, lower limit of normal.

Une femme de 40 ans vous consulte pour des paresthésies nocturnes des doigts médians de la main droite. Quel est le diagnostic le plus probable?

- a) Une thrombose de l'artère humérale
- b) Une hypersensibilité au froid
- c) Une hernie cervicale
- d) Une neuropathie cubitale au coude
- e) Une neuropathie ulnaire au poignet**

Carpal Tunnel Syndrome: A Review of the Recent Literature

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Abstract: Carpal Tunnel Syndrome (CTS) remains a puzzling and disabling condition present in 3.8% of the general population. CTS is the most well-known and frequent form of median nerve entrapment, and accounts for 90% of all entrapment neuropathies. This review aims to provide an overview of this common condition, with an emphasis on the pathophysiology involved in CTS. The clinical presentation and risk factors associated with CTS are discussed in this paper. Also, the various methods of diagnosis are explored; including nerve conduction studies, ultrasound, and magnetic resonance imaging.

Keywords: Carpal tunnel syndrome, median nerve, entrapment neuropathy, pathophysiology, diagnosis.

<http://www.ncbi.nlm.nih.gov/pmc/articles/PMC3314870/pdf/TOORTHJ-6-69.pdf>

Un homme de 63 ans, hypertendu, présente un déficit gauche à prédominance brachiofaciale, une hyposensibilité gauche et une hémiparésie gauche. Quelle est la question cruciale?

- a) Durée de l'hypertension?
- b) Autres facteurs de risque vasculaire?
- c) Est-il alcoolique?
- d) Signe de Babinski?
- e) Céphalées?
- f) **Quelle est l'heure de début? Time is brain, thrombolyse < 4:30**